

Multiple Events:

Q Describe how to calculate probabilities of multiple events.

Questions.Notes①
②

Finding the sample space of two independent events.

If two ~~events~~ independent events occur, the sample space could be listed as a table.

eg Draw a sample space showing the outcomes when a coin and die are thrown.

		Die					
		1	2	3	4	5	6
Coin	H	1H	2H	3H	4H	5H	6H
	T	1T	2T	3T	4T	5T	6T

③

How many outcomes?

The number of entries in the table

12 outcomes

How can this be used to calculate probabilities?

$$\Pr(\text{Event}) = \frac{\text{No. of outcomes event occurs}}{\text{No. of outcomes}}$$

eg

$$\Pr(\text{getting } 2T) = \frac{1}{12} \quad \begin{array}{l} \text{— number of outcomes } 2T \\ \text{— Total no. of outcomes} \end{array}$$

Summary:-

To find the total number of outcomes list the sample space in a table, Then use the table to find the number of outcomes for the event divide this by the total number of outcomes.

(6)